

Project Design Phase

Solution Architecture

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Name:	Sakhamuri Ruthvick Sai
Project Name:	Automated Network Request Management System
Maximum Marks:	4 Marks

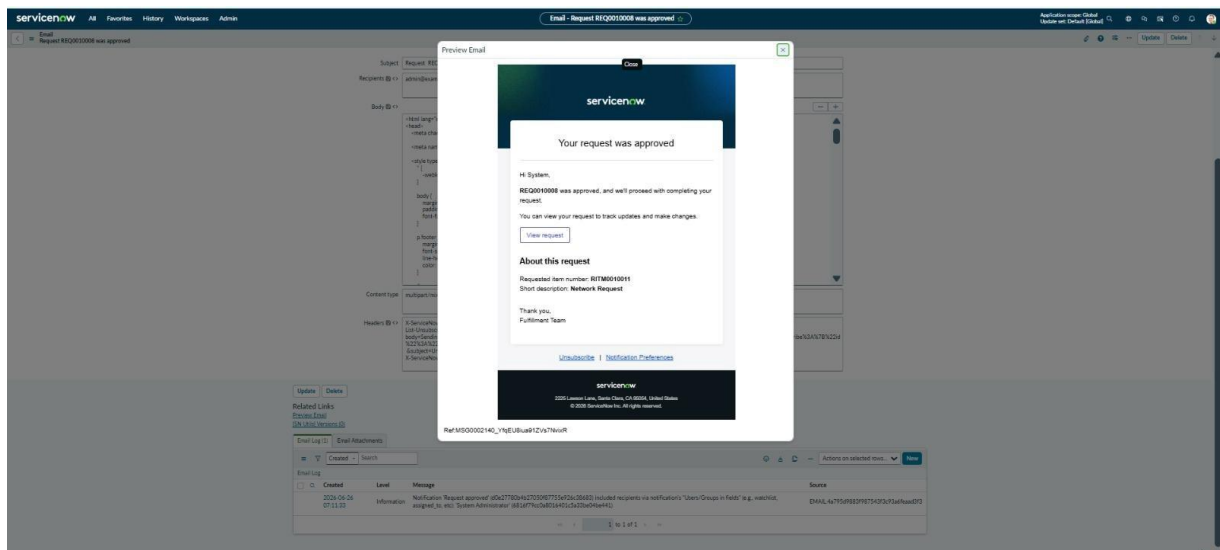
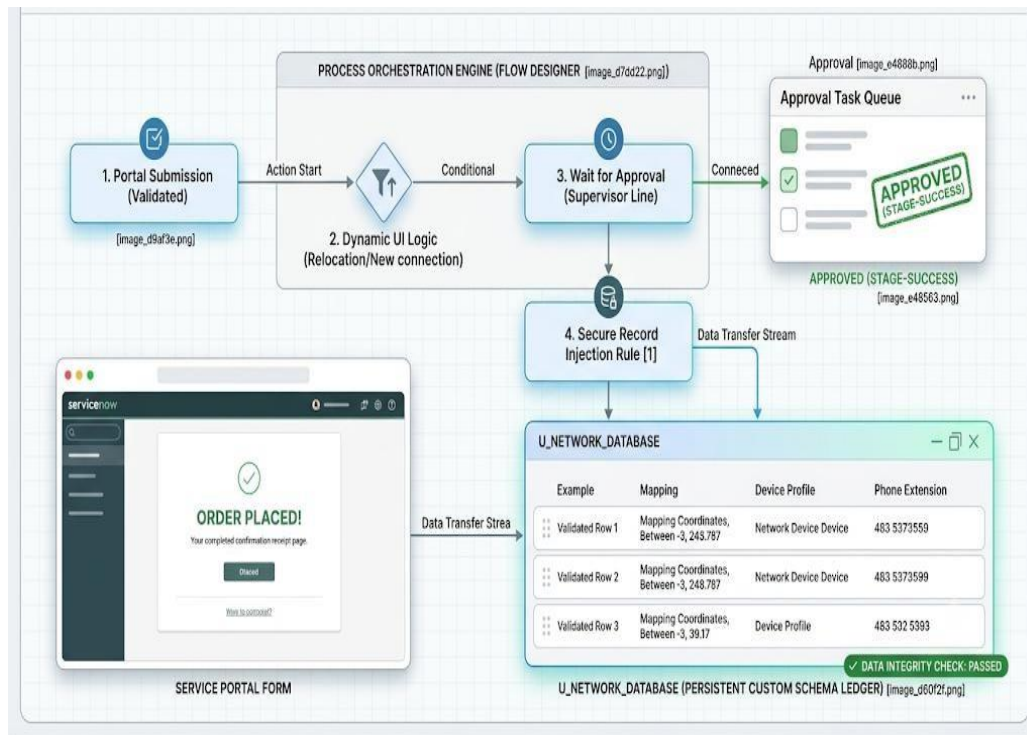
Solution Architecture Description

Solution architecture is the foundational framework that bridges enterprise business friction and automated cloud technology solutions. For this system, the architecture maps out a highly secure, transaction-validated, and transparent data path. Its core structural goals include:

- **Solving Existing Business Friction:** Eliminating manual email routing delays, clerical copy-paste bottlenecks, and coordination anxieties during department space relocations.
- **Providing Stakeholder Clarity:** Mapping exact data relationships, UI enforcement filters, approval gates, and data storage boundaries for clear engineering reference.
- **Defining System Requirements:** Outlining the execution timeline across frontend web portal objects, backend low-code workflow studio components, and secure relational database states.
- **Enforcing Technical Standards:** Giving exact specifications regarding session validation, security access control layers, data integrity checks, and direct persistence pipelines.

Solution Architecture and Data Flow Diagram

Below is the end-to-end cloud-native architecture mapping how user intake form inputs transform into structured, transaction-verified operational records within the persistent ledger.



Architectural Data Flow Breakdown

The solution architecture utilizes a multi-tier pipeline organized into four logical runtime phases to process, audit, and commit incoming network provisioning transactions:

1. Portal Submission (Validated Front-End UI Layer)

The system lifecycle initiates when an employee accesses the unified Service Portal form. To eliminate manual data entry, the interface uses client-side session logic to instantly capture the requester's core user identity context. Built-in layout rules dynamically manage field visibility, hiding or displaying complex geographical and hardware parameters based entirely on the selected transaction type (e.g., *Relocation* versus *New Connection*).

2. Process Orchestration Engine (Cloud Workflow Studio)

Once a verified form is submitted, a runtime action instantly hands the payload over to the platform's **Flow Designer Engine**. This cloud layer acts as the centralized operational controller, decoupling basic frontend entries from the backend database layer to guarantee system scalability and protect baseline system runtime availability.

3. Multi-Stage Approval Gate (Supervisor Task Queue)

The orchestration engine evaluates transaction properties against organizational routing criteria and creates an active entry inside the targeted manager's **Approval Task Queue**. The request sits at this operational milestone while the platform generates and sends real-time system notifications and email links. This ensures that no database write occurs without authorized, authenticated sign-off.

4. Secure Record Injection Rule (Data Integrity Ledger Storage)

Upon a successful approval action, the system moves into its final execution state. Instead of placing generic logs into common system tables, a dedicated server-side injection script triggers an automated data integrity sanity check. Once passed, clean row records are written directly into the custom enterprise relational database schema:

u_network_database. The portal form then instantly updates the user's interface with an "Order Placed" completion receipt page.